

# Yannam C Chiranjeevi

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## Objective

In pursuit of demanding roles as an MLOps Engineer/CUDA Engineer

## Summary

Experienced Backend Engineer and Platform Engineer with 13+ years of expertise spanning backend development, algorithmic trading, and Kubernetes platform engineering. Over the past **2+ years**, specialized in **CUDA development**, focusing on high-performance computing, GPU acceleration, and AI/ML infrastructure. Strong background in C++ programming, parallel computing, and **CUDA kernel optimizations**, with hands-on experience in **profiling, debugging, and optimizing AI workloads** on NVIDIA GPUs.

- **CUDA Development & Optimization:** Implemented custom CUDA kernels, optimized MIG/MPS configurations, and improved GPU utilization using Nsight Compute, Nsight Systems, and DCGM.
- **C++ & Backend Development:** Extensive experience in high-performance C++ programming, backend services, and optimizing algorithms for low-latency applications.
- **Kubernetes & AI Infrastructure:** Managed GPU-accelerated Kubernetes deployments, integrating GPU Operator, TensorRT, Flyte, and Kubeflow for scalable AI workloads.

## Professional Experience

July 2024 – present

**Senior Platform Engineer, QpiAI**, Bengaluru, India

Head of Infrastructure (aka Compute Service) at QpiAI

QpiAI's Quantum Division is a pioneer in South Asia, launching the region's first quantum computer, one of only seven startups selected by the Indian government's National Quantum Mission. The AI division offers a range of AI products with BYOC SaaS offerings.

- ❖ Designed, implemented, and maintained on-prem and cloud infrastructure (Azure, GCP, AWS).
- ❖ Developed CUDA kernels to optimize custom ML workloads, improving inference efficiency on A100/V100 GPUs.
- ❖ Optimized MIG/MPS configurations for TensorFlow/PyTorch workloads, leveraging Kueue-based job scheduling to improve GPU cluster efficiency by 35-45%.
- ❖ Used CUDA profiling tools (Nsight Compute, Nsight Systems) to analyze and mitigate:
  - o Memory bandwidth saturation in multi-tenant GPU workloads.
  - o PCIe bus congestion during high-volume data transfers.
  - o Suboptimal kernel launch patterns and inefficient memory allocations.
  - o CPU-GPU synchronization overhead in PyTorch/TensorFlow pipelines.
- ❖ Integrated CUDA-aware ClickHouse analytics using DCGM and Arrow for GPU performance monitoring.

- ❖ Managed PowerEdge clusters with DPUs, using SR-IOV for efficient network and storage resource allocation.
- ❖ Led the bare-metal Kubernetes deployment, integrating Rook Ceph for scalable storage and managing high-performance databases (ClickHouse, ScyllaDB, PostgreSQL)
- ❖ Contributed to FrostDB, a nascent Golang-based database focused on profiling data storage.

#### **Technologies Used:**

- ❖ CUDA & GPU Programming: CUDA, NVCC, CUDA-GDB, RAFT, cuDNN, cuDF, Nsight Compute, Nsight Systems.
- ❖ Kubernetes & Infrastructure: Kubernetes, Rook Ceph, ClickHouse, ScyllaDB, PostgreSQL, Flyte, Kubeflow, Kueue.

August 2023 – May 2024

**Staff Software Development Engineer, initializ Labs**, [Remote] Tirupati, India  
*Pepsico USA Operations, Platform Engineering Operations for K3s IoT Project*

Conducted research, offered recommendations, conducted benchmarking, ensured observability, and managed the implementation and automation of k3s infrastructure, integrating NVidia GPU, GPU Operator, and Multi-Instance GPU capabilities.

- ❖ Designed and automated K3s infrastructure with CUDA programming and NVIDIA GPU integration.
- ❖ Developed custom CUDA kernels to accelerate machine learning workloads on Kubernetes clusters.
- ❖ Configured and optimized GPU Operator for MIG-enabled workloads, improving multi-tenancy performance.
- ❖ Provided infrastructure recommendations beyond enterprise decisions, including MetalLB over kube-vip.
- ❖ Developed Ansible Playbooks for K3s provisioning on VMware machines.
- ❖ Defined a multi-tenancy strategy for Rancher Fleet, optimizing Kubernetes deployments.
- ❖ Provided observability recommendations for control plane logging and monitoring using ElasticSearch, Filebeat, Metricbeat, Prometheus, and Grafana.
- ❖ Conducted benchmarking for different K3s architectures, leveraging k-bench, Sonobuoy, and k6.

#### **Technologies Used:**

- ❖ CUDA & GPU Programming: CUDA, NVCC, CUDA-GDB, RAFT, cuDNN, cuDF, Nsight Compute, Nsight Systems.
- ❖ Kubernetes & Infrastructure: K3s, Rancher Fleet, NVidia GPU Operator, MIG, MetalLB, Ansible, Prometheus, Grafana, ElasticSearch, k-bench, Sonobuoy, k6.

November 2021 – July 2023

**Chief Technology Officer, Vincisive Technologies**, Bengaluru, India

Vincisive Technologies provides enterprise database migrations, ERP implementations, and cloud-native infrastructure solutions.

#### **Key Responsibilities:**

- ❖ Led C++ and Golang software development teams, overseeing ERP system integrations and cloud-native transformations.
- ❖ Implemented client requirements using Go Gin, Go gRPC, and C++, ensuring high-performance and scalable application development.
- ❖ Spearheaded ERPNext migration from VMs to Kubernetes, modernizing infrastructure for improved scalability and maintainability.
- ❖ Developed C++ and Golang-based containerized applications, enabling seamless cloud deployment.
- ❖ Managed 50+ Rancher clusters, integrating FluxCD GitOps for automated deployments and enhanced CI/CD pipelines.
- ❖ Assisted developers in deploying and troubleshooting C++ and Golang applications on Kubernetes-based environments.

### Technologies Used:

C++, Golang, Kubernetes, Rancher 2.x, FluxCD, Go Gin, gRPC

### Summary:

Led the **modernization of ERP systems**, transitioning to **Kubernetes-based workloads**, while overseeing **C++ and Golang software development**.

January 2019 – October 2021

#### Senior Product Engineer, **ESTATE ONE**, Bengaluru, India

- ❖ Estate One is a Product Startup in the Real Estate space
- ❖ Understanding Requirements from the Clients and the Marketing Team through Event Storming
- ❖ The Engineering Team is a Team of Backend Engineers and Self-Serving DevOps, who build, deploy and manage applications on AWS EKS and other Services end-to-end
- ❖ As a Backend Engineer, I Architected, Developed, and Deployed the Applications on AWS EKS in Microservices Architecture
- ❖ As a DevOps, I Designed and Built CI/CD Pipelines using Jenkins and FluxCD
- ❖ Managed the Infrastructure and Services such as Cassandra, using Terraform and Ansible
- ❖ Technologies used: Clojure, Ring, Pedestal, Go, Gin, Go gRPC, Jenkins, Tekton, FluxCD, Ansible, Terraform, Cassandra, Swagger (OpenAPI)

July 2018 – December 2018

#### Principal Software Engineer, **FORMCEPT**, Bengaluru, India

- ❖ FORMCEPT is a startup with flagship product MECBOT, which is a Data Ingestion, Data Management, and Data Analytics platform, which can ingest data from any type of source, unify data from various sources, and make data research-ready. The platform can also be used as a data pipeline management tool
- ❖ Part of the Engineering R&D team
- ❖ Maintaining and Making Enhancements to our flagship product: MECBOT
- ❖ Managing client projects from POC to onboarding
- ❖ NoSQL used: Cassandra, Arango; SQL: Postgres
- ❖ Message Broker/Data Ingestion: Kafka
- ❖ Stream Processor used: Spark; Elasticsearch for Analytics

- ❖ Docker/Kubernetes for Deployment, Jenkins for CI/CD, Jeera for Issue Tracking, Git for code version control

January 2017 - June 2018

**Co-founder, SR-Quant, Mumbai, India**

- ❖ We built a prototype for the Enterprise Financial Risk Technology Platform for Financial Institutes like Commercial Banks, NBFCs, and Cooperative Banks based in India
- ❖ The Enterprise Risk Technology Platform aims to mitigate financial risks that are exposed to the Financial Institutes, and Fraud Detection
- ❖ Met Key Stakeholders of many NBFCs and Banks to refine the prototype

March 2013 – December 2016

**Associate – Strategy, Estee Advisors, Gurgaon, India**

- ❖ Estee Advisors is a Hedge Fund with Algorithmic Trading Setup (Estee has been a pioneer in Algo Trading in India since 2008 i.e. since SEBI allowed Algo Trading in India)
- ❖ Developed Data Platform which ingests data from Bloomberg Professional API and other data sources (like NSE website) using C++, F#, C#, MS SQL Server, and Python
- ❖ Developed Trading Platform for the team, on top of in-house DMA platform, using C++, C# and F#
- ❖ Designed, backtested, developed, and managed Investment Products based on the clients' risk appetite and RoI needs
- ❖ Carried Investment Research combining quantitative techniques (including Machine Learning and some AI techniques) with Fundamental Research, developing factor models

April 2012 – Feb 2013

**Algorithmic Developer, Higher Technology Trading Systems (HTTS), Kochi, India**

Worked for SGFC (parent company of HTTS), a hedge-fund company based in Geneva, Switzerland

- ❖ Backtesting and validating Stat Arb strategies and Strategies based on Technical Analysis (Investment Research)
- ❖ Developed an in-house Trading Platform for Traders at HTTS and SGFC, using C++ and C#

January 2007 – June 2011

**Research Analyst, RR Donnelley (erstwhile OfficeTiger), Chennai, India**

- ❖ Worked in the Research and Analytics wing of RRD at Chennai
- ❖ Carried Financial Research and Financial Modelling in various teams working for primarily Investment Banks clients in the US and for IBM (for their Market Intelligence)
- ❖ Carried both primary research (for Mintel, London) and secondary research (for many IBs)
- ❖ Worked with Bloomberg and various financial databases like Reuters, Thomson, LexisNexis, and tools like MS Excel, VBA

April 2003 – June 2005

**Project Engineer, Silent Sentinels Electrical Consultants, Mumbai, India**

- Involved in SCADA-based power automation projects
- Taking care of all the operations at the substation level
- Communicating progress with field officers, senior management

## Education

2005–2007

**PGDBM (equivalent to MBA)** with specialization in Finance  
Institute for Technology and Management, Chennai, India

1999–2003

**Bachelor of Technology** in Electrical and Electronics Engineering  
Jawaharlal Nehru Technological University, Hyderabad, India

## Professional Certifications

**Professional Cloud Architect** from Google Cloud Platform (GCP), Certified since September 2021

**Certified Kubernetes Administrator (CKA)** from Cloud Native Computing Foundation (CNCF), Certified since May 2021

**Professional Risk Manager (PRM)** from Professional Risk Managers' International Association (PRMIA) [Certification in Financial Risk Management], USA, since June 2010